

A Beginner's Guide to Data Science and Analytics

What is Data?

Information: Data represents information, which can be structured or unstructured, quantitative or qualitative, and can be used for decision-making, analysis, or other purposes.

Alternatively, we can say that data is information, or facts, collected, stored, and used for various purposes. It can come in many forms, such as numbers, text, images, audio, or video.

Data and Its Forms:

- **When we talk about numbers, we are looking at:** Sales figures, stock prices, temperatures, house prices and many others etc.
 - **For example:** A retail store might collect numerical data on the number of items sold each day.
- **When we take Text Data:** we can examine Customer Reviews, Articles, and Documents, etc.
 - **For instance:** A company might collect customer feedback in text format through surveys or online reviews.
- **Images:** Photographs (Pictures), x-rays, and satellite imagery, etc.
 - **Example:** A health clinic might use images in the form of X-rays to diagnose bone fractures.
- **Audio:** Recordings, music, voice notes, etc.
 - **Example:** A call center might record customer calls for quality assurance and training.

- Video: Surveillance footage, movies, live streams, etc.
 - **Example:** A security agency might use video footage from surveillance cameras to monitor public spaces for safety.

Today, Data is the new oil

Today, Data is the new oil and it is the foundation of various disciplines; most importantly data science, and data analytics as they aim to transform raw data into valuable insights and thus find actionable or implementable solutions to business problems.

The phrase "Data is the new oil" suggests that data, like oil, has become a highly valuable resource in the modern world. Just as oil powered the industrial revolution and contributed to the advancement of economies, data is now driving the information age.

- **For instance,** social media companies like **Facebook and Twitter** collect vast amounts of data on user preferences and behavior. They use this data to tailor advertising to specific segments of their user base, which in turn drives revenue.

How Data is Generated in the Real World?

1. Business Transactions:

Every time a customer makes a purchase, data is generated. This can include information such as the customer's name, the items purchased, the amount spent, and the time of the transaction.

Financial Transactions: Financial institutions record data on every financial transaction, including credit card purchases, ATM withdrawals, and stock market trades.

- **Example:** In supermarkets, every time a product is scanned at the checkout, data about the sale is recorded. This data can help the store understand which products are selling well, manage inventory, and tailor promotions.

2. User-Generated Content (social media):

Social media platforms are a significant source of data. Every post, like, share, and comment generates data that can be analyzed.

Example: A brand might monitor mentions of its products on Twitter. Analyzing this data can help the brand understand public perception and sentiment, and also respond to customer feedback or complaints in real-time.

3. Sensors and Monitoring Systems:

With the advent of the Internet of Things (IoT), there has been a massive production of sensors and devices that collect data. These sensors can be found in various settings, from industrial machines to home appliances.

- **Example:** In agriculture, sensors can be used to monitor soil moisture levels. This data can be used to automatically control irrigation systems, ensuring that crops receive the right amount of water and optimizing water usage.
(<https://www.manxtechgroup.com/soil-monitoring-with-iot-smart-agriculture/>)

4. Additional Example - Wearable Devices:

People around the world increasingly use wearable devices such as smartwatches and fitness trackers. These devices continuously collect data on various aspects like heart rate, steps taken, sleep patterns, etc.

- **Example:** A person using a fitness tracker can monitor their daily physical activity and set goals. The data collected by the tracker can help them make informed decisions about their exercise routines and health.

Bellabeat, established in 2014, is a high-tech manufacturer centered around designing and developing health-focused products specifically designed for women.

The major products the company offers include the Bellabeat App, the Leaf, Time and the Spring. These products help women track and understand various aspects of their health, including activity levels, sleep, stress, and menstrual cycles. **The Leaf**, which is a versatile wellness tracker, and **Time**, a smart wellness watch, both integrate with the app to track activity, sleep, and stress. **The Spring, which** is a smart water bottle, monitors daily hydration levels.

These devices continuously collect data on various aspects like heart rate, steps taken, sleep patterns, and many others.

People generate data through various ways while using smart devices like Bellabeat's Leaf, Time, Spring, and the Bellabeat App. Here are the primary means:

1. **Activity Tracking:** Devices such as the Leaf and Time track physical activities. This includes steps taken, distance covered, calories burned, and exercise routines. This data is generated by the user's movements and activities.
2. **Sleep Monitoring:** These devices often include features that monitor sleep patterns. They track how long and how well a user sleeps, including data on deep sleep, light sleep, and periods of wakefulness.
3. **Stress Monitoring:** By tracking physiological indicators such as heart rate variability, the devices can estimate stress levels. Users generate this data through the natural variations in heart rate that occur throughout the day.
4. **Menstrual Cycle Tracking:** Through the Bellabeat App, women can input data regarding their menstrual cycles. This includes information on cycle length, symptoms, and mood changes.

5. **Mindfulness and Meditation:** The app may include mindfulness and meditation exercises. Usage data, such as frequency and duration of these exercises, can be generated.
6. **Hydration Tracking:** With the Spring water bottle, data is generated every time a user consumes water. The bottle tracks the amount of water consumed and times during the day when the user hydrates.
7. **User Inputs and Preferences:** Users may also generate data through manual inputs. This includes setting goals, preferences, or inputting data that cannot be automatically tracked, such as daily water intake (if not using Spring) or calories consumed.
8. **Social Interaction:** If the app has social features, users can generate data through interaction with other users, such as messages, shares, or participation in challenges.
9. **Device Syncing and Usage:** Data is also generated through how often users sync their devices with the app, the frequency of app usage, and which features within the app they access most frequently.

This generated data is invaluable as it helps in creating a profile of the user's health and wellness habits, which can then be used to offer personalized recommendations and insights.

Demand for Data-literate Professionals.

Yet, access to data isn't all it takes to set a business on the path to success; it also takes employees who understand and know how to leverage data. There's now an increased demand for data-literate business professionals who can handle, analyze, and interpret data to drive decision-making.

According to the U.S. Bureau of Labor Statistics, the employment of computer and information research scientists is projected to grow 21 percent from 2021 to 2031, much faster than the average for all occupations.

That is why we can see an increased demand for data-literate business professionals who can handle, analyze, and interpret data to drive decision-making.

Difference Between Data Science and Data Analytics

Now, if you're new to the world of data, two terms you're likely to encounter are "data science" and "data analytics". While these terms are related, they refer to different things. In this part we will look at an overview of what each term means and how it applies to business.

We are going to explore the differences between Data Science and Data Analytics under the following five (5) main evaluation techniques: Scope, Objectives, Techniques, Applications, and Skillset.

Data Analytics Use Case

Attaining operational efficiency and productivity for building a material company

Use Case Evaluation

In the provided use case, a prominent building material company was facing difficulties in managing and analyzing data effectively, which was critical to its operations. Their existing reporting system was inadequate, as it failed to deliver accurate results that could be used to inform decisions and strategies.

To address this problem, the company adopted an analytical solution that enabled them to keep track of essential KPIs (Key Performance Indicators) using various analytical tools. The utilization of Power BI was central to this solution. Through custom Power BI reports, the company could now visually display data on dashboards, enabling a clearer understanding of various

performance metrics. These included sales by region, team performance, frequency of orders, and information on top clients. Other technologies used in this solution included SQL for database queries, MS-Excel for data handling, and Pervasive for data management and integration.

As a result of implementing this solution, the company was able to enhance its sales productivity significantly. Moreover, the system aided in the identification of relevant data sources and enabled the development of an easy-to-navigate and visually appealing reporting system. In addition to improving efficiency and productivity, the solution also bolstered the security of the company's data.

In summary, through the implementation of a data analytics solution incorporating Power BI, SQL, MS-Excel, and Pervasive, the building material company was able to transform its data management and analysis processes. This not only enhanced operational efficiency and productivity but also improved the security and visualization aspects of their data, facilitating more informed and effective decision-making.